

CSEA20 Final

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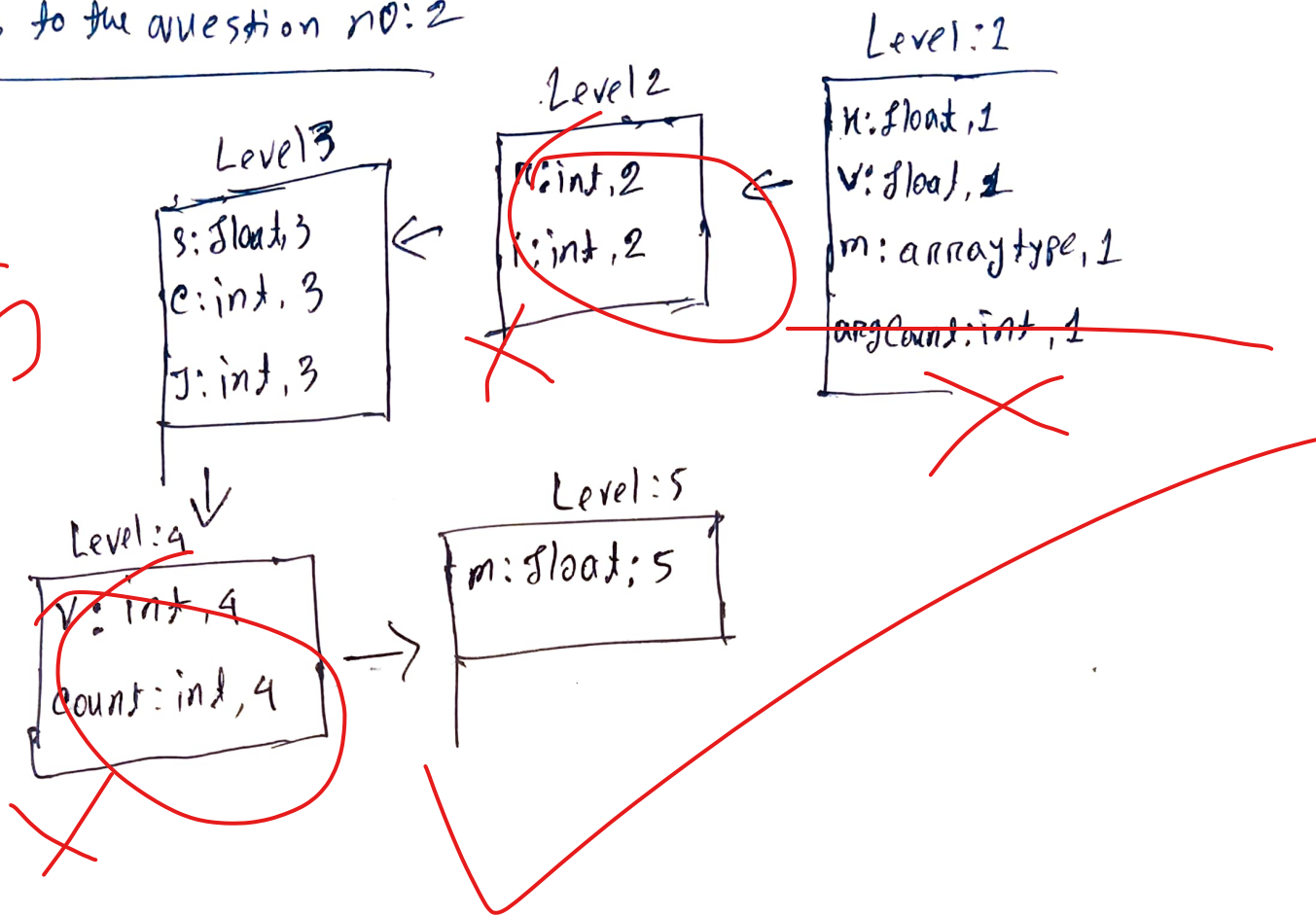
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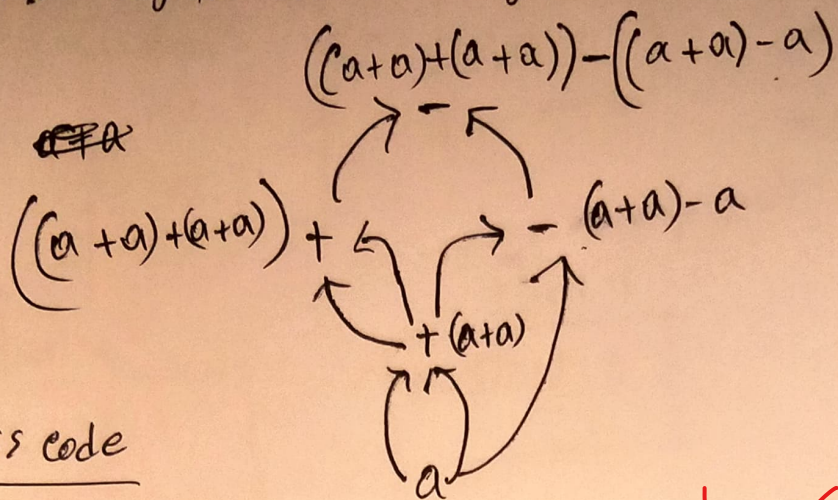
Ans to the question no: 2

03



Ans: to the question no: 3

From the acyclic graph we get the following equation



3 address code

$$((a+a)+(a+a)) - ((a+a)-a)$$

0  $t_1 = a + a$   
 1  $t_2 = t_1 - a$   
 2  $t_3 = t_1 + t_1$   
 3  $t_4 = t_3 - t_2$

Quadruples

|   | op | arg1  | arg2  | result |
|---|----|-------|-------|--------|
| 0 | +  | a     | a     | $t_1$  |
| 1 | -  | $t_1$ | a     | $t_2$  |
| 2 | +  | $t_1$ | $t_1$ | $t_3$  |
| 3 | -  | $t_3$ | $t_2$ | $t_4$  |

Triples

|   | op | arg1 | arg2 |
|---|----|------|------|
| 0 | +  | a    | a    |
| 1 | -  | 0    | a    |
| 2 | +  | 0    | 0    |
| 3 | -  | 2    | 1    |